

Theoretical modelling of torque requirements for single screw feeders

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Abstract

In the design or selection of a screw feeder the torque requirement is a principal parameter which is related to the feeder loads, properties of the bulk solid and constructional features of the screw. In this paper the load which is imposed on a screw feeder by the bulk solid in the hopper is assumed to be the flow load determined on the basis of the major consolidation stress. Five boundaries around the bulk material within a pitch are considered and forces acting on these surfaces are analysed. Particular attention is paid to the pressure distribution on the lower region of the screw. An analytical solution for the calculation of torque is determined, which allows the torque characteristics of screw feeders to be predicted. Experimental studies on the required torque for screw feeders are also reported. Two types of material, three troughs with different inside diameter and two screws with different configurations have been investigated. The results from the experiments are presented and compared with the theoretical predictions.

Keywords: Torque characteristics; Screw feeders

1. Introduction

The screw feeder is one of the most useful feeding devices which not only has good metering characteristics, but also uses relatively simple components and can be designed to feed many kinds of bulk solids reliably in a variety of applications. In the design of a screw feeder the torque requirement is an important parameter which is related to the feeder load, properties of the bulk solid and configuration of the screw.

Metcalf [1] considered the mechanics of a screw feeder, concentrating on the rate of delivery and the torque required to feed different types of coal using mining drill rods as screws. The model chosen was that of a rigid plug of material moving in a helix at an angle to the screw axis. A detailed experimental investigation was conducted by Burkhardt [2]. The tests included the effect, on the performance of a screw feeder, of the pitch, the radial clearance between the screw flight and trough, the hopper exposure and the head of the bulk solid contained in a hopper. Carleton et al. [3] discussed both screw conveyors and feeders, but from the experimental apparatus and results described in their paper there was more emphasis on screw conveyors rather than on screw feeders. Rautenbach and Schumacher [4] carried out scale-up experiments with two geometrically similar screws. By dimensional analysis the relevant set of dimensional numbers was

derived for the calculation of power consumption and capacity. More recently, Roberts and Manjunath [5] analysed the mechanics of screw feeder performance in relation to the bulk solid draw-down characteristics in the feed hopper. In their study the force exerted on the screw flights is assumed to be uniformly distributed along the whole feed length and three empirical pressure ratios are used in the determination of the required torque.

In this paper, five boundaries around the bulk material within a pitch are considered and the forces acting on these boundaries are analysed. Particular attention is paid to the pressure distribution on the lower region of the screw. An analytical solution for calculation of torque is determined which allows the torque characteristics of screw feeders to be predicted. Experimental studies on the required torque are reported.

2. Feeder loads

A typical form of hopper with a screw feeder is shown in Fig. 1. The load which is exerted on a feeder by the bulk solids in a hopper was discussed in Refs. [6–9]. There are two main loading conditions: the initial filling condition when the bin is filled from the empty state and the flow condition when discharge has occurred. The experimental evidence of this study suggests that the feeder load on a screw feeder can

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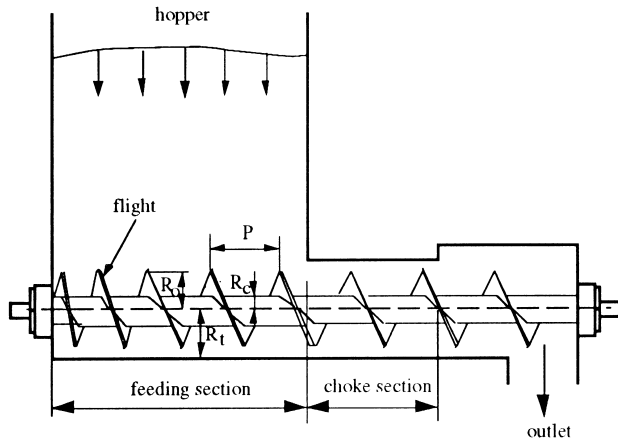


Fig. 1. Typical form of a hopper fitted with a screw feeder.

be considered to be that arising under flow conditions. Carson [10] also noted that the starting torque is close to the running torque for many bulk materials and situations, but warned that there would be exceptions, for example: bulk materials which adhere to surfaces with storage at rest (effectively increasing wall friction angles); bulk materials which gain strength with storage and require increased additional shear forces to commence flow; bins or hoppers which are vibrated during storage at rest (greatly increasing the vertical stress, i.e. feeder load). The recommended flow loads can be obtained by using methods proposed by Reisner and Eisenhart Rothe [11].

According to McLean and Arnold [6], the feeder load Q acting at the outlet of the hopper is given by

$$Q = q\gamma L^{1-m} B^{m+2} \quad (1)$$

q is a non-dimensional surcharge factor; m is a hopper shape factor: $m = 1$ for axisymmetric flow or a conical hopper, $m = 0$ for plane flow or a wedge-shaped hopper.

The flow loads on screw feeders can be determined on the basis of the major consolidation stress σ_1 . The non-dimensional flow surcharge factor may be expressed as

$$q_{f\sigma_1} = \left(\frac{\pi}{4}\right)^m \frac{Y(1 + \sin \delta)}{2(X - 1) \sin \alpha} \quad (2)$$

where

$$\beta = \frac{1}{2} \left[\phi_h + \sin^{-1} \left(\frac{\sin \phi_h}{\sin \delta} \right) \right] \quad (3)$$

$$X = \frac{2^m \sin \delta}{1 - \sin \delta} \left[\frac{\sin(2\beta + \alpha)}{\sin \alpha} + 1 \right] \quad (4)$$

$$Y = \left[\{2[1 - \cos(\beta + \alpha)]\}^m (\beta + \alpha)^{1-m} \sin \alpha + \sin \beta \sin^{1+m}(\beta + \alpha) \right] \times [(1 - \sin \delta) \sin^{2+m}(\beta + \alpha)]^{-1} \quad (5)$$

In Eq. (5) both β and α must be in radians for the term $(\beta + \alpha)^{1-m}$.

For a hopper fitted with a screw feeder, in general, $m = 0$. Based on the flow case, the feeder load can be written as

$$Q = q_{f\sigma_1} \gamma L B^2 \quad (6)$$

It is reasonable to assume that the feeder load is uniformly distributed over the hopper outlet. The resulting stress σ_o can be obtained by

$$\sigma_o = \frac{Q}{LB} = q_{f\sigma_1} \gamma B \quad (7)$$

3. Pressure on surfaces of bulk material

3.1. The five boundary surfaces

Considering the bulk material boundary in a pitch, 'pressures' are imposed on five surfaces, as indicated in Fig. 2(a). Taking account of the boundary conditions applying to the bulk material moving within screw flights, two basic regions can be specified: an upper region in which a 'shear surface' exists between the bulk solid surrounding the screw and the bulk solid propelled by the screw and a lower region in which the bulk solid is moving within a limited space which comprises rigid surfaces, as shown in Fig. 2(b). The surfaces to which pressure is applied are:

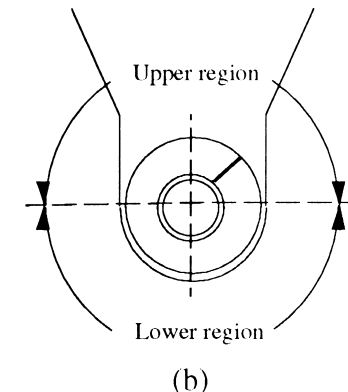
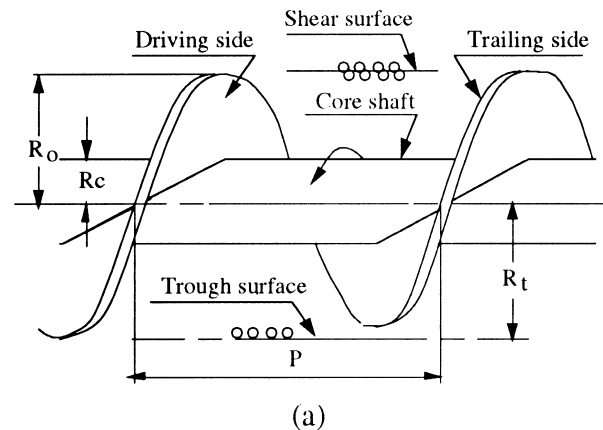


Fig. 2. Bulk material boundary within a pitch: (a) five boundary surfaces; (b) two basic regions.

- the ‘shear surface’ on the upper region of the screw;
- the trailing side of the screw flight;
- the driving side of the screw flight;
- the outside surface of the core shaft;
- the inside surface of the trough.

3.2. Pressure distribution on bulk material in the lower region

Consider the bulk material axial cross-section in the lower region in a pitch, as depicted in Fig. 3. The bulk material boundary in this region is composed of four sides: the trailing side and the driving side of the screw flight, the inside surface of the trough and the outside surface of the shaft. This boundary can be assumed to be rectangular in shape and of unit thickness. Considering the forces acting on the bulk material limited by this boundary, the force due to gravity is neglected. Because the speed of rotation is relatively low, the centrifugal effects are also regarded as negligible.

Stress σ_w is the normal wall pressure acting perpendicularly to the wall of the trough and the core shaft. σ_x is the axial compression stress. The ratio

$$\lambda_s = \sigma_w / \sigma_x \quad (8)$$

is known as the stress ratio of the bulk material sliding on the confining surfaces (i.e. trough and core shaft surfaces). A general expression can be obtained:

$$\lambda_s = \frac{\sigma_w}{\sigma_x} = \frac{1}{1 + 2\mu_d^2 + 2[(1 + \mu_d^2)(\mu_d^2 - \mu_w^2)]^{1/2}} \quad (9)$$

$\mu_d = \tan \delta$ and δ is the effective angle of internal friction of the bulk solid. μ_w is the friction coefficient between the bulk solid and a confining surface. The derivation of Eq. (9) is presented in Appendix A. For free flowing materials, $\mu_d = \mu$, where μ is the friction coefficient of the bulk solid. Then, Eq. (9) has the same form as that given by Hong and Ling [12].

When a moving bulk solid reaches steady state, there is equilibrium between the driving force and the resisting force. Assuming the axial stress and the radial stress are functions of x only, as shown in Fig. 3, the balance of forces acting on the element of length dx results in

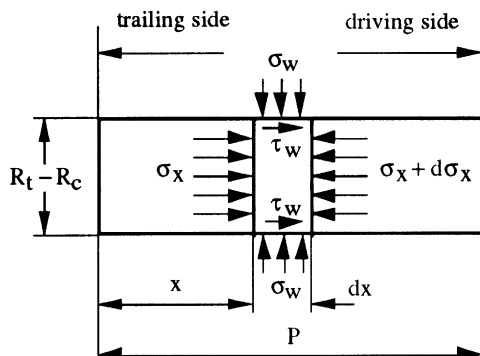


Fig. 3. Stress on an element at the lower region of the screw.

$$\frac{2\mu_w\lambda_s}{R_t - R_c} \sigma_x - \frac{d\sigma_x}{dx} = 0 \quad (10)$$

A solution to Eq. (10) is

$$\sigma_x = c_1 \exp\left(\frac{2\mu_w\lambda_s}{R_t - R_c} x\right) \quad (11)$$

where c_1 is an integration constant. For the purposes of this analysis, c_1 is determined by making the following simplifying assumption concerning the boundary condition:

$$\sigma_x = \sigma_o \quad \text{at } x = 0$$

σ_o is the stress exerted on the screw feeder by the bulk solid in a hopper. The solution to Eq. (10) can then be written as

$$\sigma_x = \sigma_o \exp\left(\frac{2\mu_w\lambda_s}{R_t - R_c} x\right) \quad (12)$$

To simplify the calculation an average radial stress along a pitch is introduced as

$$\sigma_{wa} = \frac{\lambda_s}{P} \int_0^P \sigma_x dx \quad (13)$$

Substituting σ_x from Eq. (12), the average radial stress may be derived by

$$\sigma_{wa} = \sigma_o \frac{R_t - R_c}{2\mu_w P} \left[\exp\left(\frac{2\mu_w\lambda_s P}{R_t - R_c}\right) - 1 \right] \quad (14)$$

For convenience, non-dimensional parameters are introduced:

$$c_d = d/D = R_c/R_o$$

$$c_p = P/D$$

$$c_t = (D + 2c)/D = 2R_t/D$$

Eq. (14) can be expressed as

$$\sigma_{wa} = \sigma_o \frac{c_t - c_d}{4\mu_w c_p} \left[\exp\left(\frac{4\mu_w\lambda_s c_p}{c_t - c_d}\right) - 1 \right] \quad (15)$$

In an actual application the wall friction between the bulk solid and the core shaft and the trough surface may be the same. If they are not identical, then the wall friction coefficient between the bulk solid and the trough surface can be chosen to replace μ_w , as the trough surface is dominant.

3.3. Forces acting on individual surfaces

The analysis of the forces acting upon the bulk solid element on the individual surfaces where non-cohesive bulk material is transported in a vertical screw conveyor was made by Nilsson [13]. For screw feeders the surfaces upon which forces are exerted and the status of the acting forces should be distinguished from those produced in vertical screw conveyors.

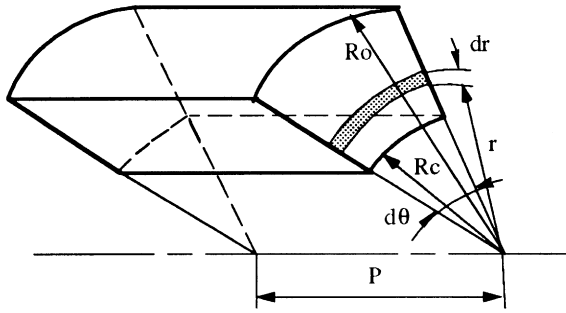


Fig. 4. A material sector in a pitch.

A bulk material sector in a pitch is used for the calculation of the axial forces acting on an individual surface, as depicted in Fig. 4.

It is assumed that within the length of a pitch the forces acting on individual surfaces are uniformly distributed (on the upper shear surface, trailing side and driving side of a flight) or these forces can be represented by average forces (on the outside surface of the core shaft and the inside surface of the trough).

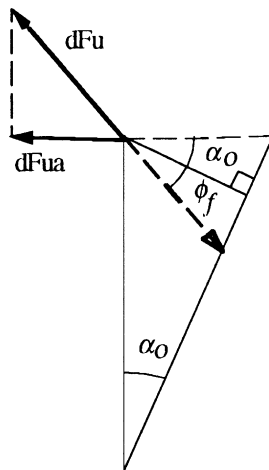
4. Axial resisting forces

4.1. Axial resisting force on the shear surface

The axial resisting force acting on the element of the bulk solid on the shear surface, as shown in Fig. 5, is given by

$$dF_{ua} = \mu_e \sigma_o R_o P d\theta \cos(\alpha_o + \phi_f) \quad (16)$$

α_o is the helical angle of the flight at the outside radius. μ_e is the equivalent friction coefficient. According to Roberts et al. [7–9], $\mu_e = (0.8–1.0) \sin \delta$.



Notation: $\dashrightarrow$ Direction of material element moving on surface
 $\longrightarrow$ Direction of resisting force acting on surface

Fig. 5. Forces on the shear surface.

The total axial force acting over a pitch length of the screw is

$$F_{ua} = \mu_e \sigma_o R_o P \cos(\alpha_o + \phi_f) \int_0^\pi d\theta \quad (17)$$

After integration and by means of non-dimensional parameters, Eq. (17) can be written as

$$F_{ua} = \frac{\pi}{2} \mu_e c_p \cos(\alpha_o + \phi_f) \sigma_o D^2 = k_u \sigma_o D^2 \quad (18)$$

and

$$k_u = \frac{\pi}{2} \mu_e c_p \cos(\alpha_o + \phi_f) \quad (19)$$

4.2. Axial resisting force on the core shaft

As shown in Fig. 6, the axial resisting force acting on the element of the bulk solid is

$$dF_{ca} = \mu_w \sigma_{wa} R_c P d\theta \sin \alpha_c \quad (20)$$

μ_w is the wall friction coefficient between the bulk solid and the core shaft. α_c is the helical angle of the flight at the core shaft.

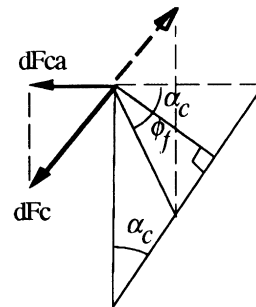
The total axial force acting over a pitch length of the screw is

$$F_{ca} = \mu_w \sigma_{wa} R_c P \sin \alpha_c \int_0^{2\pi} d\theta \quad (21)$$

σ_{wa} can be obtained from Eq. (15). Integrating Eq. (21) leads to

$$F_{ca} = \frac{\pi c_d (c_t - c_d)}{4} \sin \alpha_c \left[\exp \left(\frac{4 \mu_w \lambda_s c_p}{c_t - c_d} \right) - 1 \right] \sigma_o D^2 = k_c \sigma_o D^2 \quad (22)$$

and



Notation: $\dashrightarrow$ Direction of material element moving on shaft surface
 $\longrightarrow$ Direction of resisting force acting on surface

Fig. 6. Forces on the shaft surface.

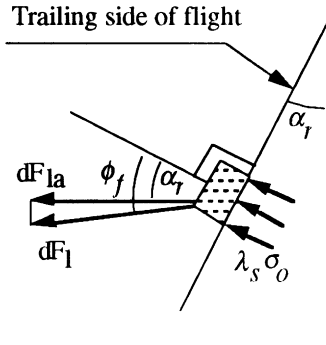


Fig. 7. Forces on the trailing side of a flight.

$$k_c = \frac{\pi c_d (c_t - c_d)}{4} \sin \alpha_c \left[\exp \left(\frac{4 \mu_w \lambda_s c_p}{c_t - c_d} \right) - 1 \right] \quad (23)$$

4.3. Axial resisting force on the trailing side of a flight

The axial resisting force acting on the trailing side of a flight as presented in Fig. 7 is

$$\begin{aligned} dF_{la} &= \lambda_s \sigma_o \frac{r dr d\theta}{\cos \alpha_r \cos \phi_f} \cos(\phi_f - \alpha_r) \\ &= \lambda_s \sigma_o r dr d\theta (1 + \tan \alpha_r \tan \phi_f) \end{aligned} \quad (24)$$

Substituting $\tan \alpha_r = P/2\pi r$ and $\tan \phi_f = \mu_f$, and integrating for r from R_c to R_o and for θ from 0 to 2π :

$$F_{la} = \lambda_s \left[\frac{\pi}{4} (1 - c_d^2) + \frac{\mu_f c_p}{2} (1 - c_d) \right] \sigma_o D^2 = k_l \sigma_o D^2 \quad (25)$$

and

$$k_l = \lambda_s \left[\frac{\pi}{4} (1 - c_d^2) + \frac{\mu_f c_p}{2} (1 - c_d) \right] \quad (26)$$

4.4. Axial resisting force on the trough surface

From Fig. 5, the axial resisting force on the trough surface is given by

$$dF_{ta} = \mu_w \sigma_{wa} R_t P d\theta \cos(\alpha_o + \phi_f) \quad (27)$$

After integrating for r from R_c to R_o and for θ from 0 to π , the total axial force over a pitch is

$$\begin{aligned} F_{ta} &= \frac{\pi}{8} c_t (c_t - c_d) \cos(\alpha_o + \phi_f) \\ &\quad \times \left[\exp \left(\frac{4 \mu_w \lambda_s c_p}{c_t - c_d} \right) - 1 \right] \sigma_o D^2 = k_t \sigma_o D^2 \end{aligned} \quad (28)$$

and

$$k_t = \frac{\pi}{8} c_t (c_t - c_d) \cos(\alpha_o + \phi_f) \left[\exp \left(\frac{4 \mu_w \lambda_s c_p}{c_t - c_d} \right) - 1 \right] \quad (29)$$

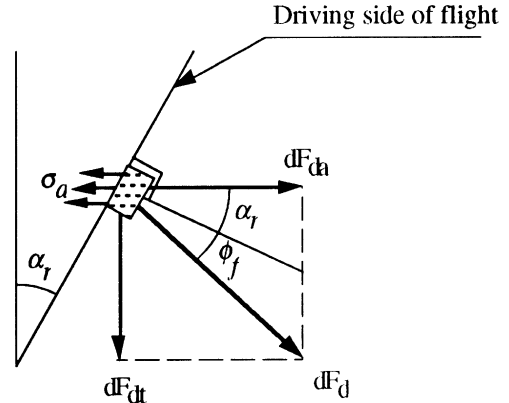


Fig. 8. Forces on the driving side of a flight.

4.5. Axial force and stress on the driving side of a flight

The axial force acting on the driving side of a flight is shown in Fig. 8. This force should be equal to the total resisting axial forces

$$F_{da} = F_{ua} + F_{ca} + F_{la} + F_{ta} \quad (30)$$

It is assumed that the total force is uniformly exerted on the surface of the driving side. The axial stress can be determined by

$$\sigma_a = \frac{F_{da}}{\pi(R_o^2 - R_c^2)} = \frac{4(k_u + k_c + k_l + k_t)}{\pi(1 - c_d^2)} \sigma_o = K_\sigma \sigma_o \quad (31)$$

where

$$K_\sigma = \frac{4(k_u + k_c + k_l + k_t)}{\pi(1 - c_d^2)} \quad (32)$$

5. Torque requirement in the feed section

Most screw conveyors can be designed with little thought given to thrust as the thrust force or axial force in an ordinary screw conveyor is moderate and commonly used screw conveyor drives will accommodate thrust in either direction. However, in a screw feeder, especially with long inlet openings, axial force can be very severe. Thus, determination of the axial force is necessary for the design of a screw feeder.

Roberts and Manjunath [5] proposed a method by which the torque requirement can be calculated based on the axial forces. An obvious advantage of this method is that both torque and axial force can be obtained from one calculation process.

Referring to Fig. 8, the tangential force on the bulk material element is

$$dF_{dt} = \sigma_a d\theta r dr \tan(\alpha_r + \phi_f) \quad (33)$$

The torque required for turning the screw is

$$T = 2\pi \sigma_a \int_{R_c}^{R_o} r^2 \tan(\alpha_r + \phi_f) dr \quad (34)$$

which becomes

$$T = K_s \sigma_a D^3 \quad (35)$$

and

$$K_s = \frac{2\pi}{D^3} \int_{R_c}^{R_o} r^2 \tan(\alpha_r + \phi_f) dr \quad (36)$$

An analytical solution to Eq. (36), by means of non-dimensional parameters, is

$$K_s = \pi c_p^3 \left[\frac{\mu_f}{12c_p^3} (1 - c_d^3) + \frac{1 + \mu_f}{8\pi c_p^2} (1 - c_d^2) + \frac{\mu_f(1 + \mu_f^2)}{4\pi^2 c_p} (1 - c_d) - \frac{3\mu_f^2(1 + \mu_f^2)}{8\pi^3} + \frac{\mu_f^2(1 + \mu_f^2)}{4\pi^3} \ln \left(\frac{\pi - \mu_f c_p}{\pi c_d - \mu_f c_p} \right) \right] \quad (37)$$

The derivation of Eq. (37) is presented in Appendix B.

Substituting σ_a from Eq. (31), the torque requirement can be expressed by

$$T = K_s K_\sigma \sigma_o D^3 \quad (38)$$

Eq. (38) is a general expression for the torque required for one pitch in the feed section. For pitch i , Eq. (38) can be written as

$$T_{fi} = K_{si} K_{\sigma i} \sigma_o D^3 \quad (39)$$

The torque required for all pitches in the feed section is

$$T_{ft} = \sum_{i=1}^{n_f} T_{fi} \quad (40)$$

where n_f is the number of pitches in the feed section.

It can be seen from Eq. (38) that the required torque is proportional to the stress exerted on the feeder by the bulk solid in the hopper and to the third power of the screw diameter. An increase of 50% in screw diameter will result in a 50% increase in opening width of the hopper and a 50% increase in the stress exerted by the bulk solid in the hopper (from Eq. (7)). According to Eq. (38) the total increase in torque will be 500%. This conclusion agrees with the dimensional analysis and experimental results reported in Ref. [4].

6. Torque calculation in the choke section

A choke section is adjacent to the hopper, as shown in Fig. 1. This section is cylindrical and has the same radial clearance as the lower part of the trough. For effective flow control the choke section should extend for at least two standard pitch lengths [14].

It is assumed that in the choke section the screw is operating 100% 'full'. The shear surface in the feed section does not exist, but is replaced by a cylindrical sliding surface. Thus, in the choke section Eq. (32) can be replaced by

$$K_{\sigma c} = \frac{4(k_c + k_i + 2k_t)}{\pi(1 - c_d^2)} \quad (41)$$

A general expression for the torque required for a pitch in the choke section is

$$T_c = K_s K_{\sigma c} \sigma_o D^3 \quad (42)$$

Normally, pitches in the choke section have the same geometry. The total torque required for pitches in the choke section can be expressed as

$$T_{ct} = n_c T_c \quad (43)$$

where n_c is the number of pitches in the choke section.

7. Torque characteristics of screw feeders

7.1. Torque components

From the analyses presented in Section 5, the torque requirement for a screw feeder is dependent on the resisting forces on the four surfaces. When the stress exerted by the bulk material in the hopper and the screw diameter are determined, the required torque can be calculated from the resisting forces acting on the individual surfaces. For the sake of convenience, the equation for the calculation of the torque requirement is rewritten as

$$T = K_s K_\sigma \sigma_o D^3$$

The factor K_σ is related to the axial resisting forces acting on the individual surfaces and is represented by the factors in Eqs. (19), (23), (26), (29) and (37). The factor K_s is derived from the tangential force acting on the driving side of a flight. To simplify the expressions, let

$$K = K_s K_\sigma = \frac{4K_s}{\pi(1 - c_d^2)} (k_u + k_c + k_i + k_t) \quad (44)$$

Let

$$K_u = \frac{4k_u K_s}{\pi(1 - c_d^2)} \quad (45)$$

K_u reflects the torque contribution from the shear surface.

$$K_c = \frac{4k_c K_s}{\pi(1 - c_d^2)} \quad (46)$$

K_c reflects the torque contribution from the core shaft surface.

$$K_i = \frac{4k_i K_s}{\pi(1 - c_d^2)} \quad (47)$$

K_i reflects the torque contribution from the trailing side of the flight.

$$K_t = \frac{4k_t K_s}{\pi(1 - c_d^2)} \quad (48)$$

K_t reflects the torque contribution from the trough surface.

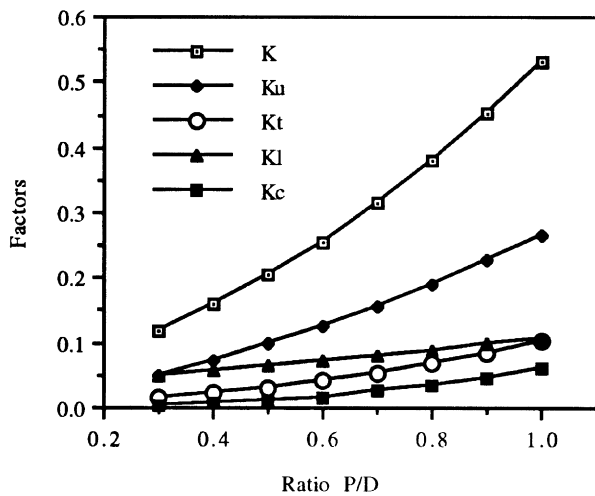


Fig. 9. Variation of factors with ratio P/D ($\mu_d=0.8$, $\mu_t=\mu_w=0.5$, $c_d=0.3$).

Fig. 9 shows the contributions of these factors to the total torque. It can be seen that the major contribution is the resisting torque acting on the upper shear surface, varying from 43% for $P/D=0.3$ to 50% for $P/D=1$. When $P/D>0.5$, factor K_u contributes 50% of the whole torque. The resisting torques on the trough surface and on the shaft surface increase with an increase of the P/D ratio, varying from 13% to 19% and from 3% to 11%, respectively. Compared with the torques on the other three surfaces, the resisting torque on the shaft surface is low, even when the value of the ratio of the core shaft diameter to screw diameter increases up to 0.5. The contribution of the resisting torque on the trailing side of the flight decreases with an increase of the P/D ratio, from 42% for $P/D=0.3$ to 20% for $P/D=1$.

7.2. Influence of clearance on the torque

The clearance between the trough and the tips of the flights is necessary to prevent metallic contact from taking place during rotation, due to various adverse factors such as deflection, minor manufacturing eccentricities, and tolerance on the screw and the trough. It is also essential to avoid nipping or wedging of particles to prevent damage and the generation of extreme contact pressure and, hence, high torques resisting rotation.

Fig. 10 gives the influence of the values of the c/D and P/D ratios on the torque requirements. The values of c/D vary from $1/64$ to $1/8$. Results from the theoretical calculation show that there is no obvious increase in factor K with a change in the ratio c/D from $1/64$ to $1/8$ for the different values of the P/D ratio. Although an increase in the clearance will result in some increase in the resisting torque (due to an increase in R_t), it also results in a decrease in radial pressure (from Eq. (12)), which feeds back to decrease the resisting torque on the trough surface and shaft surface. However, from Eq. (7), the stress exerted by the bulk solid in the hopper is linearly proportional to the opening width of the hopper outlet $B=2(D+2c)=2R_o$, which means that the stress σ_o increases

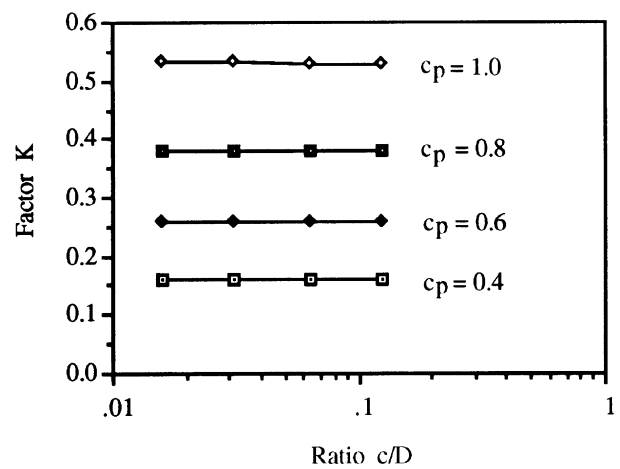


Fig. 10. Influence of c/D on K ($\mu_d=0.8$, $\mu_t=\mu_w=0.5$, $c_d=0.3$).

with an increase of the clearance. This influence of the clearance cannot be neglected in the calculation of the torque requirement.

7.3. Influence of the trough wall friction coefficient

Fig. 11 shows the effect of the trough wall friction coefficient on factor K . It can be clearly seen that K increases with an increase of the trough wall friction coefficient μ_w . Such an increase is due to two influences. First, the radial pressure along the pitch length increases with an increase in μ_w . Second, the resisting torque increases with both the radial pressure and the friction coefficient. Furthermore, the effect is stronger with a larger pitch length.

7.4. Influence of the effective angle of internal friction

The influence of the effective angle of internal friction is related to the stress ratio, shear force on the upper part of the screw and the radial pressure on the lower part of the screw. The stress ratio and the radial pressure decrease with an increase of the effective angle of internal friction, although the increase of the shear force caused by the effective angle

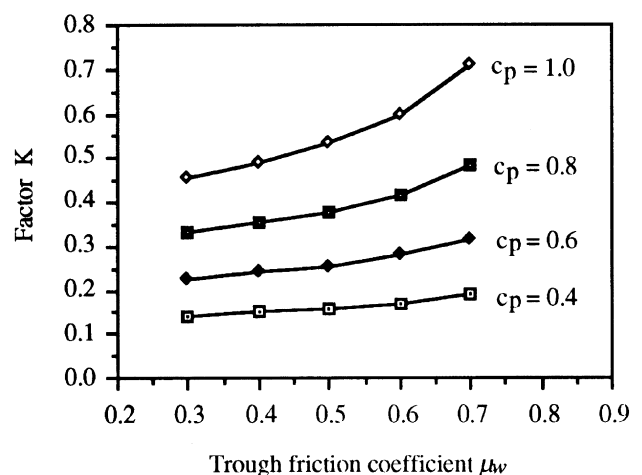
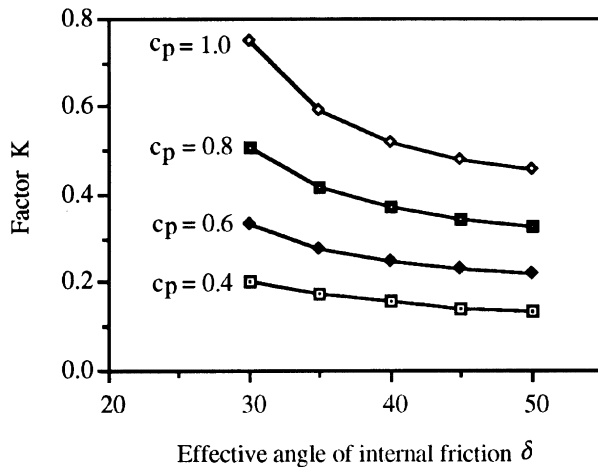
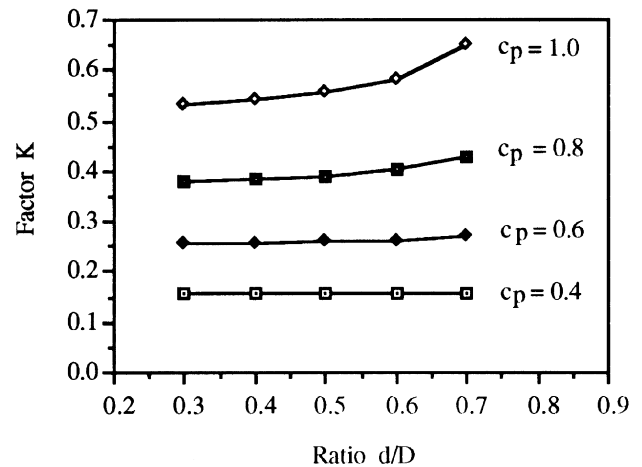
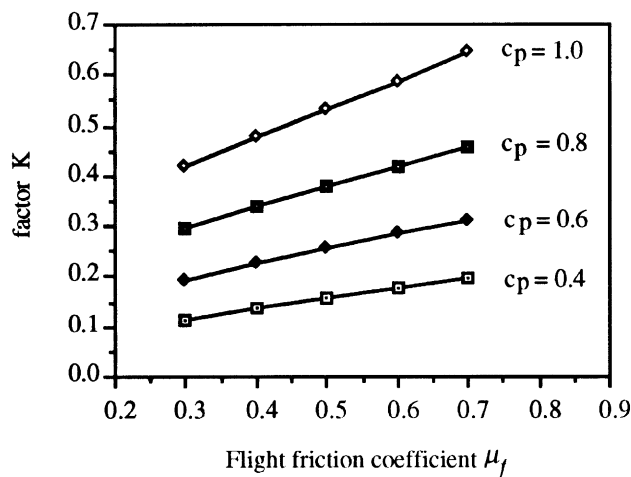


Fig. 11. Influence of μ_w on K ($\mu_d=0.8$, $\mu_t=0.5$, $c_d=0.3$).

Fig. 12. Influence of δ on K ($\mu_t = \mu_w = 0.5$, $c_d = 0.3$).Fig. 14. Influence of c_d on K ($\mu_d = 0.8$, $\mu_t = \mu_w = 0.5$).Fig. 13. Influence of μ_t on K ($\mu_d = 0.8$, $\mu_w = 0.5$, $c_d = 0.3$).

of internal friction coefficient is not sufficient to compensate for the decrease of the stress ratio and the radial pressure. Fig. 12 shows the effect of the effective angle of internal friction on the factor K .

7.5. Influence of the flight friction coefficient

Fig. 13 gives the results for K with variation of μ_f from 0.3 to 0.7. It can be seen that K becomes larger with increasing flight friction coefficient. The forces acting on both the trailing side and the driving side of the flight increase with increase of the friction coefficient between the bulk solid and the flight, resulting in an increase in the torque requirement.

7.6. Influence of the ratio d/D

The increase of c_d , the ratio of the core shaft diameter d to the screw diameter D , will result in an increase of the stress ratio, which affects the radial pressure, especially for longer pitch lengths. The resisting force and the torque on the shaft surface also increase with an increase of the ratio c_d . Fig. 14 shows the influence of the ratio c_d on K .

8. Test rig and test materials

Experimental investigations have been undertaken to understand the influence of geometric parameters, properties of the bulk solid conveyed and operating conditions on the torque required for screw feeders. A test rig, as shown in Fig. 15, was designed to monitor the performance of different types of screw (such as stepped pitch, tapered shaft and stepped shaft) in conjunction with different bulk solids, troughs and rotational speeds. This test rig consisted of five parts: hopper, trough, test screw, driving unit, receiving and weighing silo.

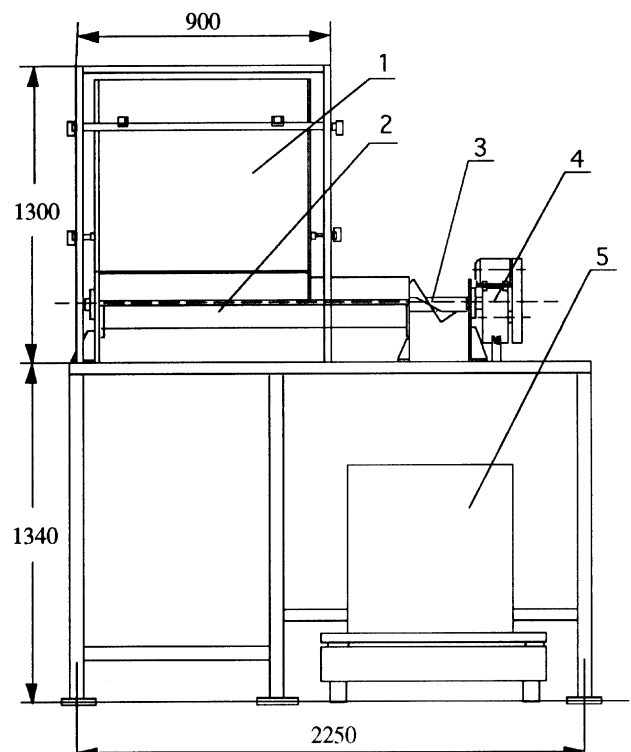


Fig. 15. Test rig for screw feeders: 1, hopper; 2, trough; 3, test screw; 4, driving unit; 5, receiving and weighing silo.

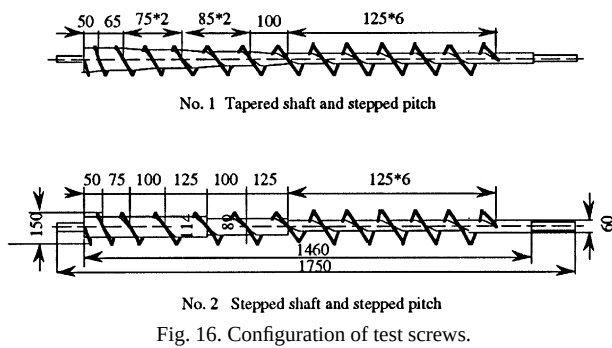


Fig. 16. Configuration of test screws.

Table 1
Physical properties of test materials

Bulk solid	d_m (mm)	ρ_b (kg m ⁻³)	δ (deg)	ϕ_w (deg)
Semolina	0.39	736	31.0	27.5
Cement	0.028	791	51.5	27.0

Screws employed for the test program had the same outside diameter ($D = 150$ mm); Fig. 16 shows the configurations of the two screws. Three troughs were used with inside diameters of 160, 170 and 190 mm, giving the radial clearances of 5, 10 and 20 mm. Two bulk solids were chosen to examine the influence of material properties on the performance of screw feeders. The main physical properties of the bulk solids have been measured and are listed in Table 1. The effective angle of internal friction and the wall frictional angle on mild steel were determined by measurements with a Jenike-type direct shear tester.

Each material was poured into the hopper to a desired level. Three different states of material in the hopper have been examined: hopper filled from an empty state to a level of 600 mm (high initial); hopper filled from an empty state to a level of 300 mm (low initial); after some discharge the level of bulk material was reduced to 300 mm (low flow).

9. Experimental results

Detailed experiments under three different filling states in the hopper and for a screw speed range of 10–80 rpm were carried out using a trough with clearance $c = 5$ mm. The experimental results did not show any obvious difference due to the different filling states and screw speeds for the test materials.

Figs. 17 and 18 give the comparison between the theoretical predictions and the experimental results for two screws with three troughs. In the keys, C is short for cement and S for semolina. The three radial clearances are 5, 10 and 20. For the tapered shaft screw (No. 1) an average value of the shaft diameter within an individual pitch has been assumed in the calculations. For screw No. 2 with a stepped diameter shaft the corresponding shaft diameter for an individual pitch has been used. It can be seen that the theoretical predictions are reasonably consistent with the experimental data, except

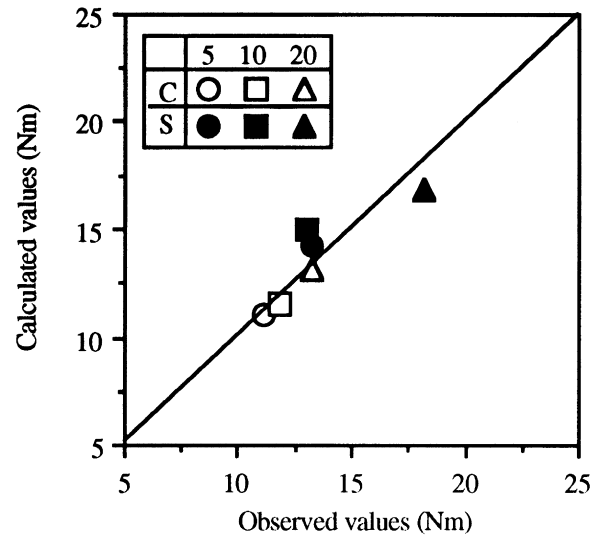


Fig. 17. Comparison of torques for a screw feeder using the screw with a tapered shaft and stepped pitches.

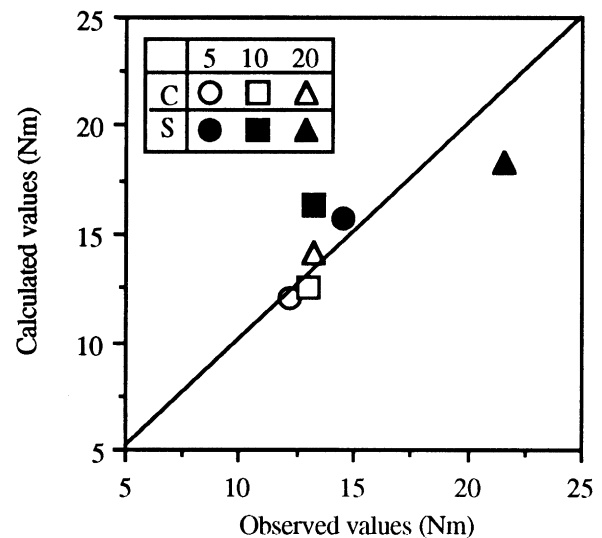


Fig. 18. Comparison of torques for a screw feeder using the screw with a stepped shaft and stepped pitches.

for the screw operating in the largest trough ($R_t = 95$ mm) with semolina, where the calculated values are lower than the observed results.

Semolina is a very free flowing material. The output of the screw feeder with semolina increases with increasing radial clearance. However, the torque requirement does not show the same pattern. It can be observed from the experimental results that for both screw feeders the torque required for the middle trough ($R_t = 85$ mm, $c = 10$ mm) is less than that for the other two troughs. There are two reasons for this situation. For bigger shaft diameters of the screws, the small clearance ($c = 5$ mm) would result in an increase in the pressure in the lower region, thereby increasing the torque requirement. On the other hand, the large radial clearance ($c = 20$ mm) may result in an additional resisting force due to the difficulty of the larger mass of bulk solids (assumed to be sliding on the

trough surface in the lower region) trying to ‘enter’ the region of the shear surface above the screw.

10. Conclusions

1. A theoretical model for the torque requirement of a screw feeder is developed by applying principles of powder mechanics to a moving material element within a pitch. The model indicates that the torque requirement is proportional to the stress exerted on the feeder by the bulk solid in the hopper and to the third power of the screw diameter.

2. Consideration of the forces acting on the five confining surfaces surrounding the bulk solid contained within a pitch, and the pressure distribution in the lower region of the screw, leads to a reasonable prediction of the torque requirement for a screw feeder.

3. The analytical solution for the calculation of the torque requirement can reveal the relationships among the screw geometry parameters, properties of the bulk solid and feeder load arising from the hopper. Thus, the torque characteristics can be better understood.

4. Based on the parameters used for this study, the torque required for a screw feeder is determined mainly by the resisting torques acting on the shear surface; the proportion of the total torque requirement is about 50%.

5. Experimental results indicate that the starting torque is close to the running torque for the test materials and situations. The feeder load exerted by bulk solid in the hopper can be determined by the flow load based on the major consolidation stress.

11. List of symbols

B	opening width of hopper outlet (m)
c	clearance between trough and tip of flight (m)
c_1	integration constant in Eq. (11)
c_d	ratio of shaft diameter to screw diameter
c_p	ratio of pitch to screw diameter
c_t	ratio of trough inside radius to screw outside radius
d	core shaft diameter (m)
d_m	mean equivalent particle diameter (m)
D	screw diameter (m)
F_{ca}	axial resisting force on core shaft surface (N)
F_{da}	axial resisting force on driving side of flight (N)
F_{dt}	tangential force on driving side of flight (N)
F_{la}	axial resisting force on trailing side of flight (N)
F_{ta}	axial resisting force on trough surface (N)
F_{ua}	axial resisting shear force on upper region of screw (N)
k_c	factor defined in Eq. (23)
k_l	factor defined in Eq. (26)
k_t	factor defined in Eq. (29)
k_u	factor defined in Eq. (19)

K	factor defined in Eq. (44)
K_c	factor defined in Eq. (46)
K_l	factor defined in Eq. (47)
K_s	factor defined in Eq. (37)
K_t	factor defined in Eq. (48)
K_u	factor defined in Eq. (45)
K_σ	factor defined in Eq. (32)
$K_{\sigma c}$	factor defined in Eq. (41)
L	length of feed section (m)
m	hopper shape factor in Eq. (1)
n_c	number of pitches in choke section
n_f	number of pitches in feed section
P	pitch length (m)
q	non-dimensional surcharge factor
$q_{f\sigma_1}$	surcharge factor for flow condition based on σ_1
Q	feeder load exerted by bulk solids in hopper (N)
r	radius of flight (m)
R_c	radius of core shaft (m)
R_o	outside radius of flight (m)
R_t	inside radius of trough (m)
T	torque required for driving screw (N m)
T_c	torque required in choke section (N m)
T_f	torque required in feed section (N m)
x	coordinate
X, Y	factors in feeder load equations, flow condition

Greek letters

α	hopper half-angle (deg)
α_c	flight helix angle at core shaft (deg)
α_o	flight helix angle at outside diameter (deg)
α_r	flight helix angle at radius r (deg)
β	angle in Eq. (3) (deg)
γ	specific weight of bulk solid (N/m ³)
δ	effective angle of internal friction of bulk solid (deg)
θ	polar coordinate
λ_s	stress ratio of bulk solid sliding on surface
μ	friction coefficient of bulk solid, $\mu = \tan \phi$
μ_d	tangent of effective angle of internal friction, $\mu_d = \tan \delta$
μ_e	equivalent friction coefficient of bulk solid, $\mu_e = (0.8-1) \sin \delta$
μ_f	wall friction coefficient between bulk solid and flight
μ_w	wall friction coefficient between bulk solid and confining surface
ρ_b	loose poured bulk density (kg/m ³)
σ_1	major consolidation stress (Pa)
σ_a	axial stress on driving side of flight (Pa)
σ_m	stress in Fig. 19 (Pa)
σ_o	stress exerted by bulk solid in hopper (Pa)
σ_w	wall stress in Fig. 3 (Pa)
σ_{wa}	average wall stress defined in Eq. (13) (Pa)
σ_x	axial stress in Fig. 3 (Pa)
τ_w	shear stress on confining surface (Pa)

$$\begin{aligned}
K_s = 2\pi \frac{P^3}{D^3} & \left\{ \frac{\mu_f}{3} \left[\left(\frac{R_o}{P} \right)^3 - \left(\frac{R_c}{P} \right)^3 \right] \right. \\
& + \frac{1 + \mu_f}{4\pi} \left[\left(\frac{R_o}{P} \right)^2 - \left(\frac{R_c}{P} \right)^2 \right] \\
& + \frac{\mu_f(1 + \mu_f^2)}{4\pi^2} \left(\frac{R_o}{P} - \frac{R_c}{P} \right) - \frac{3\mu_f^2(1 + \mu_f^2)}{16\pi^3} \\
& \left. + \frac{\mu_f^2(1 + \mu_f^2)}{8\pi^3} \ln \left(\frac{2\pi R_o/P - \mu_f}{2\pi R_c/P - \mu_f} \right) \right\} \quad (B7)
\end{aligned}$$

Employing non-dimensional parameters allows K_s to be expressed as

$$\begin{aligned}
K_s = \pi c_p^3 & \left[\frac{\mu_f}{12c_p^3} (1 - c_d^3) + \frac{1 + \mu_f}{8\pi c_p^2} (1 - c_d^2) \right. \\
& + \frac{\mu_f(1 + \mu_f^2)}{4\pi^2 c_p} (1 - c_d) - \frac{3\mu_f^2(1 + \mu_f^2)}{8\pi^3} \\
& \left. + \frac{\mu_f^2(1 + \mu_f^2)}{4\pi^3} \ln \left(\frac{\pi - \mu_f c_p}{\pi c_d - \mu_f c_p} \right) \right] \quad (B8)
\end{aligned}$$

which is Eq. (37) of the main text.

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